

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: QUANTUM, ATOMIC AND MOLECULAR PHYSICS

TRINITY TERM 2018

Wednesday, 13 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) What is the physical origin of the fine structure in hydrogen-like atoms? Show that an expression for the energy shifts is

$$\Delta E_{\text{so}} = \frac{\beta}{2} [j(j+1) - l(l+1) - s(s+1)],$$

where l and s are the orbital angular momentum and spin quantum numbers, and $\mathbf{j} = \mathbf{l} + \mathbf{s}$. Derive an expression for β .

[6]

[Hint: If $l \neq 0$, the mean inverse cubic distance of an electron is given by

$$\left\langle \frac{1}{r^3} \right\rangle = \left(\frac{Z}{na_0} \right)^3 \frac{1}{l(l+1/2)(l+1)},$$

where n is the principal quantum number, a_0 the Bohr radius and Z the nuclear charge.]

(b) Draw an energy level diagram showing the fine structure arising from the configurations of the $n = 2$ and $n = 3$ shells of the He^+ ion (a hydrogen-like system). Calculate the energy shift for each level, and list all the allowed transitions between these levels.

[6]

(c) An exotic ion is formed by the binding of an Ω^- particle to a bare Pb nucleus ($Z = 82$). The Ω^- is a particle (belonging to the baryon family) with charge $-e$ and mass $m_{\Omega} = 1672 \text{ MeV}/c^2$. The $n = 10$, $l = 9$ level is split into four separate levels. Determine the spin of the Ω^- baryon and calculate the energy separation of these levels, giving your answer in eV.

[6]

(d) Consider now the case of two Ω^- particles bound to a Pb nucleus. Label all the energy levels for the excited states in which one Ω^- particle has the principal quantum number $n = 1$ and the other one $n = 2$. What levels are possible in the ground configuration of this system? Compare your answer with the case of a regular helium atom.

[7]

2. (a) Draw a diagram showing the levels arising from the ground configuration and the first two excited configurations of the helium atom, labelling them with the appropriate spectroscopic notation and indicating all the electric-dipole allowed transitions. Which transitions are likely to be seen in absorption? Are intercombination lines likely to occur for this element?

[5]

(b) A constant magnetic field is now applied to the atoms. Discuss what is meant by the weak field Zeeman effect in the LS coupling scheme and derive an expression for the perturbation of the energy levels.

[4]

(c) Draw diagrams showing the possible emission lines arising from $(1s3s)^3S_1 \rightarrow (1s2p)^3P_1$ and $(1s3s)^1S_0 \rightarrow (1s2p)^1P_1$ transitions, indicating their energy separation and their polarization. Label your diagrams according to whether these transitions are split by the normal or the anomalous Zeeman effect.

[4]

(d) An experiment shows that a total of 6 lines are seen from these two transitions. What can you conclude about the orientation of the observer with respect to the magnetic field direction?

Assume that the dominant line broadening mechanism is due to collisions among helium atoms. If the helium gas is kept at a temperature of 10 K and a pressure of 0.1 Pa, estimate the minimum magnetic field required to resolve these lines. You can treat helium as an ideal gas consisting of solid spheres of radius 0.1 nm.

[6]

(e) The positron is the antiparticle of the electron. The electron and the positron have the same absolute magnetic moment, but have g -factors with opposite signs. Positronium is the bound system of an electron and a positron. What are the possible values of the total spin of this atom-like system in the ground configuration? What is the effect of applying a magnetic field to the ground configuration of this system?

[6]

3. (a) Consider a two level atomic system, with energies E_2 and E_1 for the upper and lower states respectively, so that $E_2 > E_1$. In the presence of an oscillating electric field from laser light the occupation amplitudes of these states, $c_1(t)$ and $c_2(t)$, are given by

$$\begin{aligned}\dot{c}_1(t) &= -i\Omega_R \cos(\omega t) e^{-i\omega_0 t} c_2(t), \\ \dot{c}_2(t) &= -i\Omega_R \cos(\omega t) e^{i\omega_0 t} c_1(t),\end{aligned}$$

with ω the laser frequency and $\omega_0 = (E_2 - E_1)/\hbar$ the resonant frequency. You may assume that Ω_R is a real number. Discuss the meaning of Ω_R and give an expression for it. [4]

(b) For the initial conditions $c_1(0) = 1$ and $c_2(0) = 0$, show that, in the rotating wave approximation, the solution of the above equations in the case of weak light is

$$|c_2(t)|^2 = \frac{\Omega_R^2}{\alpha^2} \sin^2(\alpha t/2),$$

and find an expression for α .

[Hint: in the presence of a weak perturbation the occupation amplitudes remain similar to their initial values, allowing the differential equations above to be replaced by much simpler approximations.] [7]

(c) Sketch your solution as a function of the detuning $\delta\omega = \omega - \omega_0$ for some fixed value of t . [3]

(d) Now assume that there is an additional level, with energy $E_3 > E_2 > E_1$, and that the equations describing the occupation amplitudes of the three levels are now given by

$$\begin{aligned}\dot{c}_1(t) &= -i\Omega_R \cos(\omega t) e^{-i\omega_0 t} c_2(t), \\ \dot{c}_2(t) &= -i\Omega_R \cos(\omega t) e^{i\omega_0 t} c_1(t) - i\Phi_R \cos(\omega t) e^{-i\phi_0 t} c_3(t), \\ \dot{c}_3(t) &= -i\Phi_R \cos(\omega t) e^{i\phi_0 t} c_2(t),\end{aligned}$$

where $\phi_0 = (E_3 - E_2)/\hbar$, and Φ_R is a real number. For weak light with initial condition $c_1(0) = 1$ and $c_2(0) = c_3(0) = 0$, find an approximate expression for $c_2(t)$. [3]

(e) To find an approximate expression for $c_3(t)$ assume that the laser is approximately resonant with the average of the two energy gaps, so that $\Delta\omega = \omega_0 + \phi_0 - 2\omega$ is much smaller than all the other frequency combinations in the problem. Show that the occupation probability for the third level can be written in a form similar to that given above for the second level, that is,

$$|c_3(t)|^2 = \frac{\Omega_{\text{eff}}^2}{\beta^2} \sin^2(\beta t/2),$$

and find an expression for Ω_{eff} and β . [8]

4. (a) Define the Einstein coefficients for atomic transitions, indicating which physical processes they correspond to and giving expressions for the rate at which energy is emitted or absorbed. [6]

Consider a two level system interacting with a narrow beam of intensity $I(\nu) = I\delta(\nu - \nu_L)$, where ν_L is the laser frequency. The upper and lower levels have lifetimes τ_u and τ_l , respectively. The energy separation between the two levels is $h\nu_0$, and the broadening of each level is described by a line-shape function $\phi(\nu - \nu_0)$. In steady state, the population inversion density as a function of the laser intensity is given by

$$N^*(I) = \frac{N^*(0)}{1 + I/I_S},$$

where $N^*(0) \equiv N_u - (g_u/g_l)N_l$, with N_u and N_l are the number densities of atoms in the upper and lower states, respectively, g_u and g_l are the statistical weights of the upper and lower states. The saturated intensity is given by

$$I_S = \frac{h\nu_L}{\sigma_{ul}} \left(\tau_u + \frac{g_u}{g_l}\tau_l - \frac{g_u}{g_l}\tau_u\tau_l A_{ul} \right)^{-1},$$

where A_{ul} is the Einstein coefficient for spontaneous emission, and σ_{ul} is the optical gain cross section.

(b) Assume that the frequency of the laser beam is close to that of the transition ($\nu_L \approx \nu_0$), and that the levels are homogeneously broadened with linewidth $\Delta\nu$. If the upper level is depopulated predominantly by radiative decay, prove that the saturation intensity is proportional to ν_0^3 . [5]

Consider the case of an astrophysical laser whose active medium consists of an elongated gas cloud, approximately a cylinder of diameter d and length L , with $d \ll L$, pumped by a nearby star. Lasing on the transition considered above is initiated by spontaneous emission within a region of length $\ell \ll L$. The small distance ℓ corresponds to one gain length, that is: $\sigma_{ul}N^*\ell = 1$, and the population inversion is initially large.

(c) Show that the total rate at which photons are spontaneously emitted from within this small region can be approximated by

$$\frac{\pi d^2 I_S}{4h\nu_0}. \quad [4]$$

(d) The light beam is progressively amplified as it propagates through the length of the cloud. Show that the minimum length of the cloud required to amplify the beam to saturation is given by

$$L_S \simeq \frac{I_S}{h\nu_0 A_{ul} N_u} \ln(1/\eta),$$

where η is an efficiency factor that accounts for losses in the medium (you can model the atoms in the cloud as a two level system). [7]

(e) The efficiency factor is mainly due to the fact that photons are initially emitted in all directions, instead of being directed towards the other end of the cloud. From this consideration, provide an estimate for η . [3]